Open-World Explained-Rarity Filtering for Health-Insurance Claim Fraud: Problem Formulation, Benchmark Design and Preliminary Synthetic Evidence

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***Abstract*—Health-insurance fraud detectors trained on known fraud categories fail in two ways: they force novel schemes into familiar classes (the closed-world assumption), and they treat statistical rarity as evidence of fraud, which penalizes legitimate severe cases such as polytrauma. We formulate *explained-rarity filtering*: a claim should be escalated only if it is both unfamiliar to the model and unexplained by its clinical context. We describe OW-ERF, a pipeline that (i) scores novelty by fusing an energy score and a Mahalanobis distance, (ii) converts the score into a split-conformal p-value for approximate control of the false-rejection rate, and (iii) filters rare-but-explained claims with a heteroscedastic expectation model trained on legitimate claims only. We evaluate a tabular prototype on a synthetic claim generator with four fraud mechanisms under leave-one-mechanism-out. In this preliminary setting the conformal stage keeps the benign rejection rate near the nominal 5%, and the filter reduces the benign escalation rate to about 0.5%, but it also lowers recall on mechanisms whose attributes look clinically plausible. We report this trade-off and the limitations of synthetic-only evaluation. We make no claim about real-world performance.**

***Index Terms*—health-insurance fraud, open-world recognition, conformal prediction, anomaly detection, graph neural networks**

I. Introduction

Large claim volumes in health-insurance systems, including IRDAI-regulated insurance and the AB-PMJAY scheme in India, make exhaustive manual review impractical. Fraud mechanisms such as upcoding, unbundling, phantom billing and collusive referral rings also change over time. Three problems follow.

1) **Closed-world classification.** A softmax classifier assigns a new scheme to the nearest known class, often with high confidence.
2) **Rarity is not fraud.** Pure anomaly scores flag legitimate severe cases because they are rare, complex and expensive.
3) **Alert fatigue.** Raw scores carry no statistical meaning, so a fixed threshold drifts and review budgets are exceeded.

Contributions. (1) A problem formulation, *explained-rarity filtering*, that separates unfamiliarity from unexplainedness. (2) A pipeline (OW-ERF) combining post-hoc novelty fusion, split-conformal rejection and a heteroscedastic clinical expectation filter. (3) A benchmark protocol with a synthetic claim generator, leave-one-mechanism-out evaluation and leakage control. (4) Preliminary evidence from a tabular prototype, including where the filter hurts.

II. Related Work

Open-set and open-world recognition. Open-set recognition [1], [2] and out-of-distribution (OOD) baselines [3]–[5] score unfamiliarity from logits or embeddings. On graphs, OpenWGL [6] rejects unseen-class nodes with a variational graph autoencoder, and GNNSafe [7] applies energy-based detection to graph neural networks.

Graph fraud and anomaly detection. CARE-GNN [8] counters camouflage with reinforcement-learned neighbor selection, and BWGNN [9] addresses heterophily. DOMINANT [10] and the PyGOD library [11] cover reconstruction-based detection, while BOND [12] and GADBench [13] provide benchmarks. These methods output novelty or anomaly scores and do not separate rare-but-clinically-explained claims from unexplained ones.

Conformal prediction. Split-conformal methods [14]–[16] give finite-sample marginal guarantees under exchangeability. Barber et al. [17] study relaxations, and Huang et al. [18] apply conformal prediction to graph neural networks. Temporal claim streams violate exchangeability, so our guarantee is approximate and is tested empirically.

Heteroscedastic regression. Predicting a mean and a variance under a Gaussian likelihood [19], [20] lets the expected spread grow with severity.

Healthcare fraud. Prior work applies supervised and unsupervised learning to Medicare-style data [21].

III. Problem Formulation

Let a claim v have features x_v, a graph neighborhood, and a label y_v from benign, known mechanisms, or novel mechanisms. Training sees only benign and known mechanisms. At test time a claim is *unfamiliar* if its conformal p-value satisfies $p_v \le \alpha$, and *unexplained* if its clinical inconsistency ratio a_v exceeds a calibrated threshold τ. A claim is escalated if and only if it is both.

To prevent temporal leakage, the embedding of claim v at adjudication time t uses only the causal neighborhood

$$\mathcal{N}_r(v;t) = \{u : (u,v) \in E_r,\ t(u,v) \le t\}. \tag{1}$$

IV. Method

Fig. 1 shows the pipeline.

A. Encoder

The full system builds a time-stamped heterogeneous graph (claim, beneficiary, hospital, doctor, package, diagnosis) and encodes it with R-GCN [22] or HGT [23] using PyTorch Geometric [24]. We also describe three complementary branches whose individual value must be established by ablation: an LSTM over the provider's recent claim history, a one-dimensional CNN over the ordered procedure-code sequence of a claim, and a Transformer encoder over relational-context tokens. *The repository now includes a runnable synthetic-data training pipeline for the LSTM, CNN and Transformer branches. The preliminary results below were generated by the tabular prototype only; they are not results from a trained full-model comparison or from real insurance claims.*

B. Novelty score

From classifier logits z, the energy is

$$E(x) = -T \log \sum_k \exp(z_k/T). \tag{2}$$

The Mahalanobis score $M(h)$ is the minimum over known classes of the squared Mahalanobis distance of embedding h to the class mean under the class covariance. With F_E and F_M the empirical CDFs of E and M on training data, the fused score is

$$u = \tfrac{1}{2}F_E(E) + \tfrac{1}{2}F_M(M). \tag{3}$$

C. Conformal rejection

On a calibration set disjoint from training and containing only known-class claims, with n calibration scores u_i, the p-value is

$$p_v = \frac{1 + \#\{i : u_i \ge u_v\}}{n+1}, \tag{4}$$

and a claim is rejected as unfamiliar when $p_v \le \alpha$ with $\alpha = 0.05$.

D. Explained-Rarity Filter

A heteroscedastic model trained on benign claims only predicts a mean μ_j and standard deviation σ_j for $j \in \{\text{log-amount}, \text{length of stay}\}$ from *exogenous* inputs (a severity proxy from prior history, comorbidity count, age) by minimizing

$$\mathcal{L}_g = \frac{1}{N}\sum_i \sum_j \Big[\ln \sigma_{ij} + \frac{(o_{ij}-\mu_{ij})^2}{2\sigma_{ij}^2}\Big]. \tag{5}$$

The inconsistency ratio is

$$a_v = \max_j \max\Big\{0, \frac{o_j - \mu_j}{\sigma_j}\Big\}. \tag{6}$$

The threshold τ is set as a quantile of a_v on benign calibration claims rather than by hand. The expectation model must not condition on fields a fraudster controls, such as the claimed diagnosis code; otherwise upcoding would make fraud look explained.

E. Planned components

Episodic pseudo-novel training masks one known fraud class per episode and adds a squared-hinge energy-margin loss to cross-entropy. HDBSCAN [25] groups escalated claims into typologies. Neither is evaluated here.

V. Experimental Protocol

Data. A synthetic generator produces benign claims, 5% rare-severe legitimate claims, and four mechanisms: upcoding, unbundling, phantom billing, and collusion (represented in the prototype by two proxy features only). The fraud rate is 10%.

Splits and metrics. Leave-one-mechanism-out: train and calibrate on benign plus three known mechanisms, test on all four. We use five random seeds and $9{,}000$ test claims per run. Metrics are recall on the held-out mechanism, the benign rejection rate, and the fraction of rare-severe legitimate claims flagged, each before and after the filter.

Baseline. Isolation Forest [26], with its alert count matched to the number of OW-ERF escalations.

VI. Preliminary Results

Table I reports means over five seeds from the released script. The conformal stage rejects between 4.3% and 4.6% of benign claims, close to the nominal 5% on this i.i.d. synthetic split; this does not demonstrate validity under temporal drift. After the filter, benign escalation falls to about 0.5%. The filter cuts flags on rare-severe legitimate claims most clearly for phantom billing ($13.3\% \to 2.6\%$). It also lowers recall: for collusion the drop is large ($0.586 \to 0.211$), because those claims look clinically plausible and would require graph structure that this prototype lacks. At a matched alert count, Isolation Forest recalls only 0.02–0.04 of the three attribute-level mechanisms (0.20 for collusion) while flagging 22%–28% of rare-severe legitimate claims. All of these results are conditional on the generator.

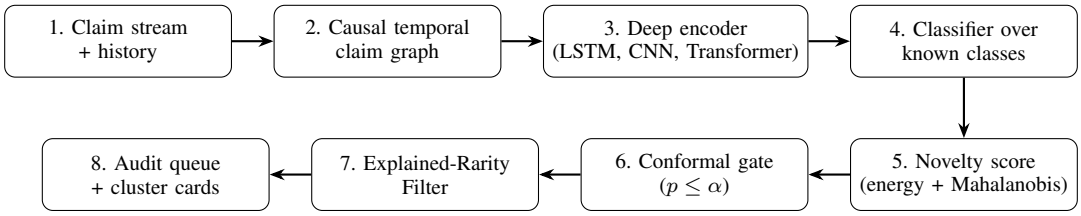


Fig. 1. Pipeline of OW-ERF. Stages 1–4 learn representations; stages 5–8 make the decision.

TABLE I
LEAVE-ONE-MECHANISM-OUT RESULTS (SYNTHETIC DATA, TABULAR PROTOTYPE, MEAN OF 5 SEEDS). RECALL IS ON THE HELD-OUT MECHANISM; "RARE FLAG" IS THE FRACTION OF RARE-SEVERE LEGITIMATE CLAIMS FLAGGED.

Held-out mechanism	Recall conf.	Recall +ERF	Rare flag conf.	Rare flag +ERF
Upcoding	0.574	0.560	0.026	0.006
Unbundling	0.493	0.372	0.020	0.003
Phantom billing	0.496	0.411	0.133	0.026
Collusion (proxy)	0.586	0.211	0.016	0.002

VII. LIMITATIONS

The evidence is synthetic only, and the generator encodes our assumptions about severity and cost. The prototype has no graph encoder, so collusion is only approximated by two tabular features. Novelty scoring is post-hoc, without episodic pseudo-novel training. The conformal guarantee is marginal and assumes exchangeability, which temporal streams violate, and it does not cover the filter stage. If the filter's severity inputs can be manipulated, fraud can be explained away. Only one baseline has been run so far. Finally, wrongly escalated claims delay care, so any deployment requires human review and an audit of group-wise error rates.

VIII. CONCLUSION AND FUTURE WORK

Explained-rarity filtering trades recall for far fewer false alarms on legitimate rare claims. Next steps are the graph encoder, trained and ablated deep branches, the full baseline set, temporal conformal tests, and evaluation on a real public claims dataset.

DATA AND CODE AVAILABILITY

No patient data are used. The synthetic generator, tabular prototype, and experiment code are available at https://github.com/ishantpant2121/MUJ-DS-2430010379. The reported results use synthetic data only; no real insurance claims are included. A versioned archival DOI may be added if a future public release is deposited.

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